

Light and Retinopathy of Prematurity: Does Retinal Location Offer a Clue?

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ABSTRACT. Nursery illumination has been implicated in the pathogenesis of retinopathy of prematurity (ROP), although the results of recent studies are conflicting. The data base for this article is a prospective ROP study on 607 infants of birth weight ≤ 1700 g including 35 larger siblings from multiple births when 1 infant fulfilled the birth weight criteria. Retinopathy commences preferentially in the nasal retina of the most immature neonate and is less likely to develop, or its onset is delayed, in the superior and inferior regions. These findings cannot be fully accounted for by regional vascular and neuroanatomical variations. Radiometric and physiological evidence suggests that the very immature neonate, most at risk of developing severe ROP, receives the greatest retinal irradiance. Furthermore, ROP commences in the areas of the retina receiving the highest light dose, and its onset is either retarded or inhibited in the darker retinal regions. Further studies are required to determine whether early exposure to light is a factor in the development of ROP. If a causal relationship is proven, here at least is one modality that can easily and immediately be controlled. *Pediatrics* 1992;89:648-653; *retinopathy of prematurity, light, natural history, preterm infant.*

ABBREVIATIONS. ROP, retinopathy of prematurity; GA, gestational age; PCA, postconceptual age.

Terry,^{1,2} in his original descriptions of retrolental fibroplasia, now called retinopathy of prematurity (ROP), first suggested that early exposure to light might be a causative factor. This was not substantiated by early studies as retinopathy still developed despite bilateral patching at birth³ or patching of one or both eyes within 24 hours of birth.⁴ Given the outcome of these studies, the prevailing view in the early 1950s^{5,6} was that early exposure to light could not be implicated in the pathogenesis of retrolental fibroplasia and that supplemental oxygen was the sole etiological factor. In 1969 Riley and Slater⁷ proposed that light could generate free radicals by damaging the retinal tissues and thus suggested another mechanism by which this condition might be caused.

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No additional clinical studies were performed until 1985, when Glass and associates⁸ reported a reduced incidence of ROP in neonates exposed to reduced (150 lux) compared with standard (600 lux) neonatal nursery illumination. This study was later criticized by Stobie⁹ for a sequential rather than concurrent design, a failure to randomize, and the use of unmasked observers. The effect of light diminution in reducing ROP has since been observed by Hommura and coworkers¹⁰ but not by Ackerman and associates.¹¹

If light is a factor in the etiology of ROP, its effects may be subtle and it could influence both the occurrence and subsequent progression of ROP. Regional variations of retinal illumination could thus affect the time of ROP onset and its retinal location. To investigate these issues, certain aspects of the natural history of this condition need to be taken into account, such as the age at onset,¹² its transient nature, and regional variations across the retina.^{13,14} To date, the possible effects of light on ROP have not been adequately considered in this context.

The purpose of this article is to explore the hypothesis that light is a contributory factor in the pathogenesis of ROP. We present evidence drawn largely from two sources of information: (1) a prospective study of the natural history of ROP in which all retinal regions were systematically examined on a regular basis; and (2) a series of studies on light and the eye of the preterm neonate which investigated the physical (environmental) and physiological factors controlling the amount of light reaching the neonatal retina. The ROP study provides the data base for this paper, whereas information on the neonatal ocular light dose will be introduced in the discussion.

PATIENTS AND METHODS

Between July 1, 1985, and May 31, 1987, a prospective study of ROP was undertaken in all five neonatal units in three cities of the East Midlands of England. All infants with birth weights of 1700 g or less who were admitted to one of these neonatal units, who survived the first 3 weeks, were enrolled. In the case of multiple birth, larger siblings were also included when one infant fulfilled the birth weight criteria. No infant received supplemental vitamin E. Risk factors¹⁵ and the epidemiological aspects of this study have been previously reported¹⁶; for the latter, a subgroup of infants born to mothers resident within defined geographical boundaries was used.

Ophthalmic examinations by a single observer (A.R.F.) commenced at 3 weeks of age and were conducted weekly while the infant was resident on the neonatal unit, every 2 weeks after hospital discharge until 12 weeks of age, and thereafter as clinically indicated. Retinal examination was done by indirect ophthalmoscopy after pupillary dilation (cyclopentolate 0.5%, with or without

phenylephrine 2.5% eyedrops). Before the age of 12 weeks, topical anesthesia (sodium benoxinate 0.4%) was instilled and an eyelid speculum and scleral indentation were used to enable the entire retinal circumference to be visualized. Findings for acute stages were recorded according to the International Classification of ROP.¹⁷ The data collected included retinal findings for each location by clock-hour and zone and were analyzed on an IBM 3090 computer using the statistical package SPSS. Medians and interquartile ranges are quoted for continuous variables and percentages for categorical variables.

Approval for this study was granted by the Leicestershire, South Derbyshire, and Nottingham District Ethical Committees.

RESULTS

The cohort consisted of 607 infants: 572 with birth weights of 1700 g or less and their 35 siblings from multiple births who exceeded this birth weight. There were 339 boys and 268 girls; their ethnic origin was as follows: white, 494 (81.4%); Indo-Pakistani, 70 (11.5%); Afrocaribbean, 26 (4.3%); and other groups, 17 (2.8%). The birth weight and gestational age (GA) distributions of the cohort are shown in Figs 1 and 2. Median birth weight was 1371 g (range 490 to 3000 g) and median GA 31 weeks (range 24 to 40 weeks). For clarification the term gestational age is not used after birth. The term postconceptual age (PCA) is

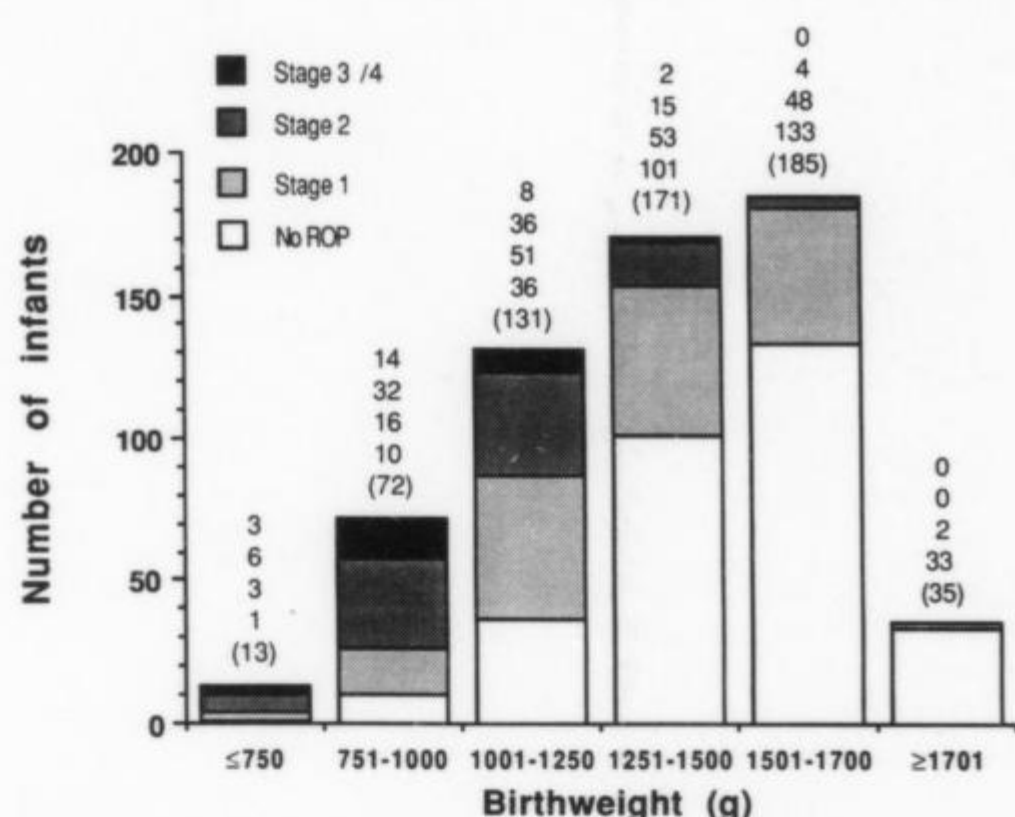


Fig 1. Birth weight distribution of cohort and maximum stage of retinopathy of prematurity (ROP). Numbers on each column indicate, from the top downward, numbers of babies with each stage of ROP. Totals in parentheses.

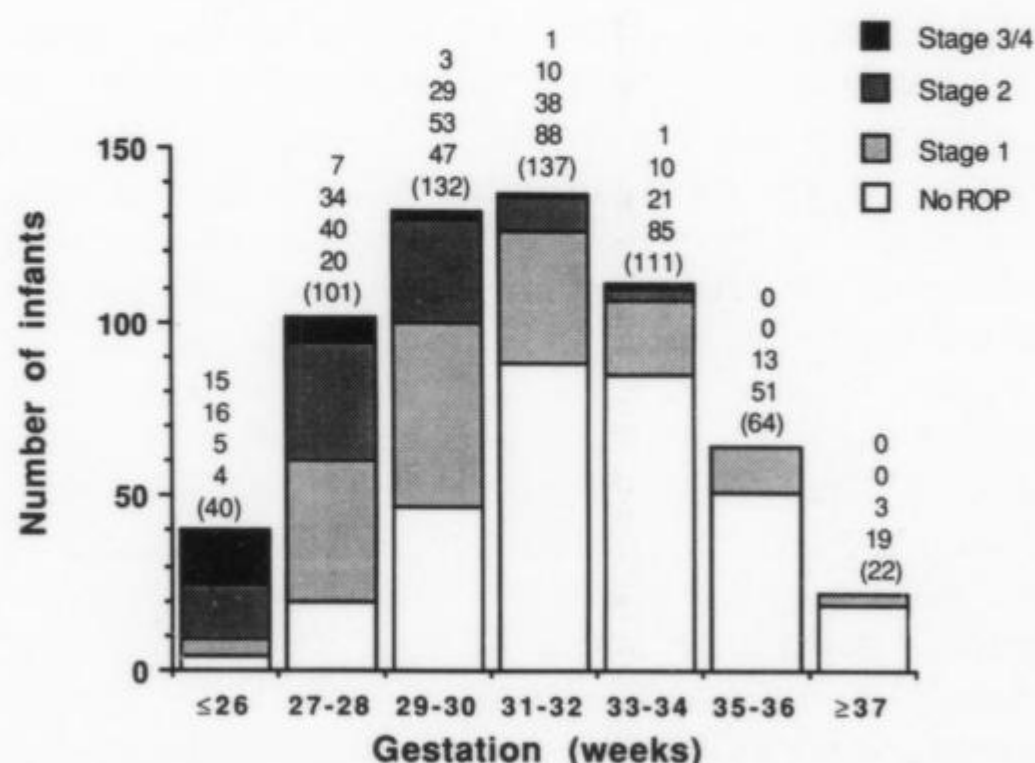


Fig 2. Gestational age distribution of cohort and maximum stage of retinopathy of prematurity (ROP). Annotations as in Fig 1.

defined as gestational age plus age from birth and is used in preference to postnatal age to describe age after birth, taking into account the degree of prematurity.

During the first 12 weeks of life there were 2865 examinations, of which a speculum and scleral indentation were used in 98.4%. During this period 24 infants died or moved away from the study area; of the remaining 583, 535 were available at 3 months corrected age. The number of examinations per baby in the first 12 weeks after birth was as follows: two or fewer, 17.7%; three through five, 47.7%; six through eight, 29.0%; nine or more, 5.6%. Acute ROP developed in 293 (48.2%) infants. The maximum stage reached was stage 1 in 173 (23.7%), stage 2 in 93 (15.1%), stage 3 in 25 (4.1%), and stage 4 in 2 (0.3%). The stage reached by birth weight and GA is shown in Tables 1 and 2. All ROP presented initially as stage 1 and if progression occurred this was through stages 2 and then 3. Interocular ROP differences were observed in 75 cases, but in none by more than 1 stage.

The location of ROP at onset is presented in Fig 3. As ROP usually involves more than one clock-hour, each eye often contributes to more than one point on the graph. Retinopathy most commonly commenced in the nasal and temporal retina and less frequently in the superior and inferior regions (Fig 3). Left- and right-eye patterns of ROP onset mirrored each other.

TABLE 1. Site and Age at Onset of Retinopathy of Prematurity (ROP): Left Eye*

Site (Clock-Hours)	No.	Median GA, (wks)	Median PCA at Onset, wk
0	335	32 (31-34)	...
3 (T)	184	30 (28-31)	35 (34-37)
6 (I)	48	29 (27-31)	35 (34-37)
9 (N)	101	28 (26-29)	33 (32-34)
12 (S)	26	27 (26-29)	34 (33-36)

* Gestational ages (GAs) of neonates according to ROP presence and site, and the postconceptual ages (PCAs) at which this developed. Onset of ROP was first seen in the nasal retina and is delayed in vertical regions. For reasons outlined in the text, data for temporal onset are less accurately timed. Numbers in parentheses represent interquartile ranges. Letters in parentheses refer to retinal regions: T, temporal; I, inferior; N, nasal; S, superior.

TABLE 2. Site and Age at onset of ROP: Right Eye*

Site (Clock-Hours)	No.	Median GA, (wk)	Median PCA at Onset, (wk)
0	343	32 (31-34)	...
3 (N)	81	29 (26-29)	33 (32-36)
6 (I)	36	29 (27-31)	35 (34-37)
9 (T)	183	30 (28-31)	35 (35-37)
12 (S)	27	27 (26-29)	34 (33-36)

* See footnote to Table 1 for explanation.

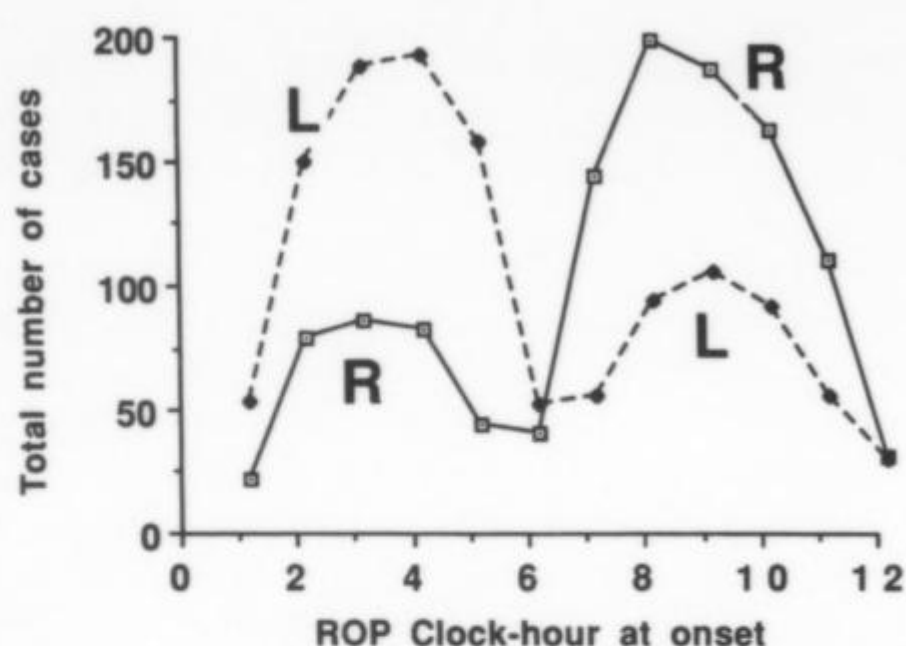


Fig 3. Location of retinopathy of prematurity (ROP) displayed by retinal clock-hour at onset. Individual neonates often contribute more than one point. Solid line, right eye; broken line, left eye.

The median GA distributions according to site of onset in left and right eyes, respectively, are shown in Tables 1 and 2. The median GA of babies according to either the presence or absence of ROP and its site of onset in, for example, the left eye, was as follows: no ROP, 32 weeks; temporal onset (3 o'clock), 30 weeks; inferior (6 o'clock), 29 weeks; nasal (9 o'clock), 28 weeks; and superior (12 o'clock), 27 weeks GA. The PCA at onset, also for the left eye, was as follows: temporal, 35 weeks; inferior, 35 weeks; nasal, 33 weeks; and superior, 34 weeks. Data for temporal onset were less accurate as these infants tended to be more mature, were often discharged from hospital before the onset of ROP, and thus received ocular examinations only every two weeks. The results indicate that, in the more immature neonate, ROP commences first in the nasal retina, and its onset in the vertical regions is delayed. The results for the right eye are similar (Table 2). Figure 4 shows the site of onset and existing or subsequent clock-hour involvement. The tendency of ROP to extend circumferentially increased as follows: temporal < inferior < nasal < superior. Thus temporal onset ROP exhibited the least, and superior the most, predilection to extend circumferentially.

Each point in Fig 5 represents the proportion of infants (expressed as a percentage) who for each clock-hour of ROP at any time also had retinopathy at that site at onset. Thus for any ROP observed at the 3-, 6-, 9-, or 12-o'clock position, the likelihood that it had been present at that site, at onset, was as follows: for the right eye, 78%, 35%, 76%, or 25%; and for the left eye, 74%, 35%, 86%, or 34%, respectively. Using the left eye as an example, for ROP observed at any time in the nasal retina (9 o'clock), there was a high likelihood (86%) that it was present at that same site at the time of onset of the process, whereas at the 12-o'clock position this was only 34%.

DISCUSSION

The natural history of ROP has been studied prospectively and the following main points have emerged. The retinal location of ROP at onset varies and is influenced by GA, and ROP has a predilection to develop first in the nasal retina of the most immature neonate.^{13,14} This finding has also been reported by Nissenkorn et al,¹⁸ who observed that stage

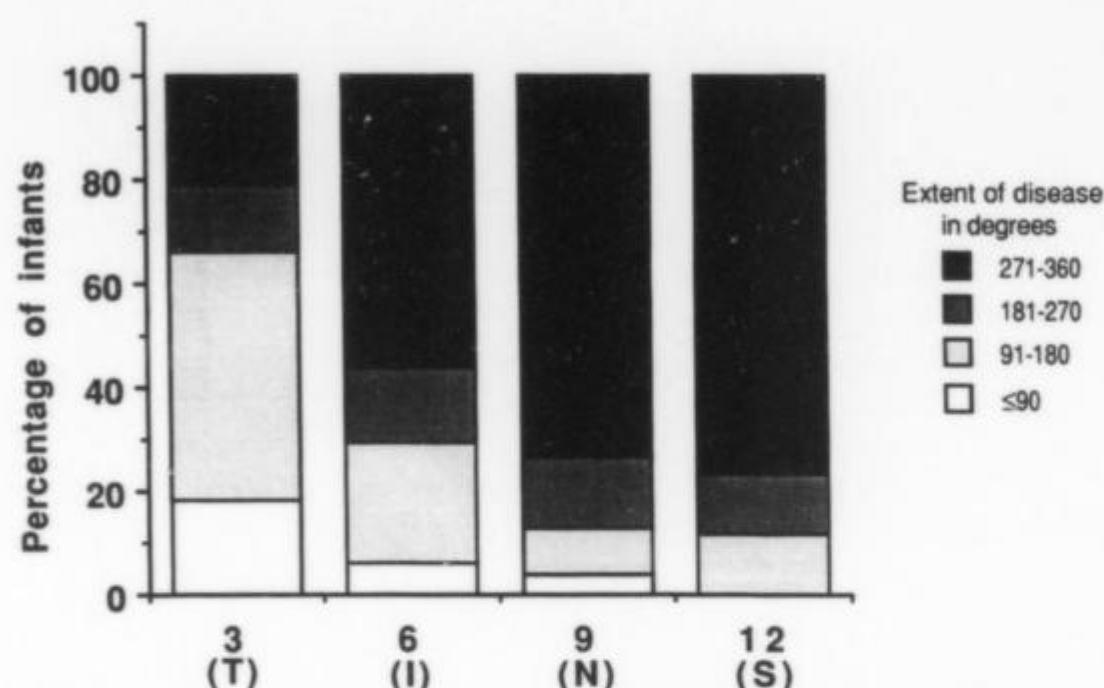


Fig 4. Site of retinopathy of prematurity (ROP) onset and clock-hour involvement: left eye. Each column represents the extent of retina involved, in degrees, at any time of the ROP process. For example, ROP commencing in superior retina has a propensity for greater clock-hour extension than retinopathy first observed in the temporal retina. Letters in parentheses refer to temporal (T), inferior (I), nasal (N), and superior (S) retinal regions.

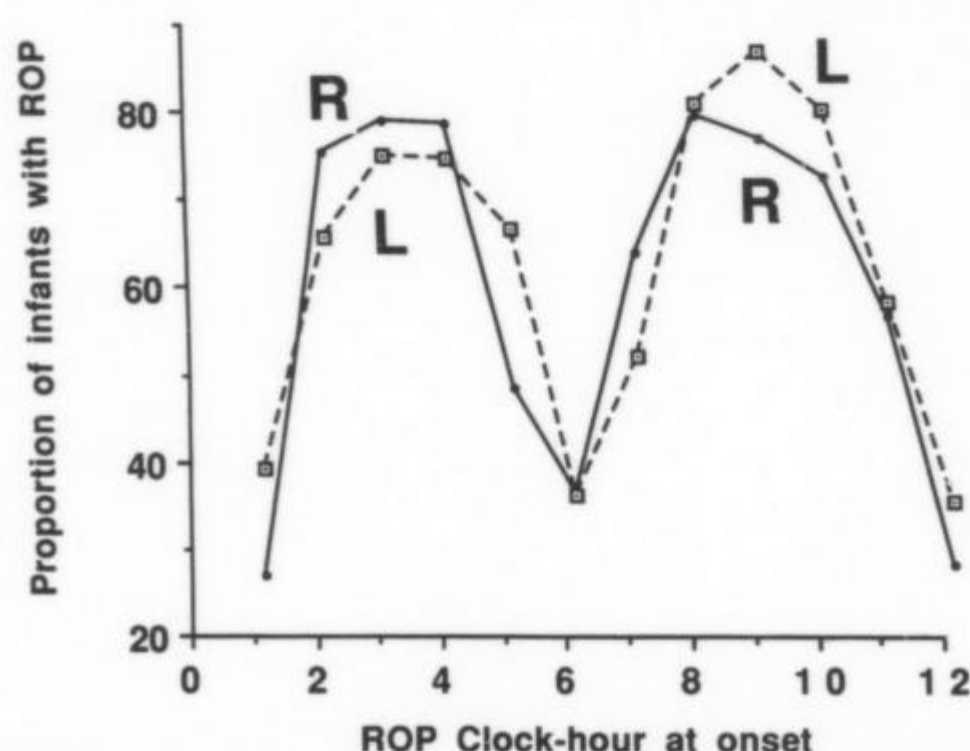


Fig 5. Proportion of neonates, as a percentage, who for each retinal clock-hour of retinopathy of prematurity (ROP) at any time also had retinopathy at that site at onset. For example, in the left eye, for ROP present at any time in the nasal retina (9-o'clock position), there was a high probability (86%) that it was present at that same time at the onset of the process. In contrast, at the 12-o'clock position this likelihood was only 34%. Solid line, right eye; broken line, left eye.

3 developed 2 weeks earlier in the nasal compared with the temporal retina. That it is less likely to develop in the superior and inferior retinal regions has not been previously reported, and if they are involved, its onset at these sites is frequently delayed. In an attempt to explain these findings the discussion will focus on three aspects: retinal vascular development, retinal neural development, and regional variations of the retinal light dose. Considering the large number of observations, a degree of observer error cannot be entirely excluded; however, two factors mitigate against this being a major contributor to our findings. First, the study protocol was specifically designed to investigate the subtle features of the natural history of acute ROP. Examinations commenced early, were performed frequently, and by scrutinizing the entire retinal circumference, both time of onset and clock-hour location could be determined as accurately as clinically possible. Second, the

consistent pattern of our findings was mirrored in each eye.

Retinal vascular development commences with the centrifugal growth of mesenchymal cells from the vicinity of the hyaloid artery at the optic disc, from around the 15th and 16th weeks of gestation. Patient vessels are formed from the posterior border of this advancing mesenchymal vanguard.¹⁹ At 5 months GA vessels extend only 2 mm from the optic disc and by 7 to 8 months GA have reached the ora serrata nasally and the equator temporally,²⁰ this latter region being vascularized in all by the 10th month GA.²¹ Michaelson²² observed at all ages the following order of vessel length: upper temporal > lower temporal > nasal. These studies¹⁹⁻²³ make no comment on the vascular development in the vertical retinal regions.

Retinopathy of prematurity affects only the immature retinal vasculature and develops at the growing ends of the developing retinal vessels: at the junction of the vascularized and avascular retina. This explains its temporal onset in relatively mature infants whose retina is vascularized except in that region. However, in the very immature neonate with a wide avascular zone involving the full extent of the peripheral retina, the ROP process does not commence simultaneously across the entire circumference, but has a predilection to involve first the nasal retina. In our study, prior completion of vascularization prevented the development of retinopathy in the vertical regions in some; however, this was not so for all of our infants as ROP did sometimes occur in these areas, but its onset was frequently delayed (see Fig 3, Tables 1 and 2). Thus, while the degree and pattern of retinal vascular development determine the occurrence of ROP and its location in the relatively mature neonate, it cannot fully account for the regional variations reported here in the very immature neonate.

It is not appropriate to consider retinal neuroanatomical regional variations, as such factors could influence local metabolic requirements and hence promote ROP. In the adult, photoreceptor densities decline toward the periphery, but not uniformly in all directions. The density of retinal photoreceptors in the human^{24,25} and monkey,²⁶ and ganglion cells in the human^{27,28} and monkey,²⁶ is higher in the nasal compared with the temporal retina, elongating the isodensity contours horizontally for both cones and rods. This asymmetry extends into the far periphery. Photoreceptor density also declines with increasing eccentricity along the vertical meridia but the asymmetry between superior and inferior regions is less pronounced than horizontally. Inferiorly, the density of cones is slightly greater, and for rods it is slightly lower compared with the superior retina.²⁵ Provis et al,²⁸ in the only study on prenatal eyes, detected nasotemporal asymmetry even in the first trimester. Two further aspects need also to be considered. First, retinal vascular and neural organization is not concentric, the former being centered on the optic disc and the latter on the fovea. This eccentricity (15 degrees), by displacing the vascular pattern nasally, reduces the nasotemporal asymmetry from the viewpoint of cell density, although the magnitude of this effect has not been quantified. Second, the period

before term is one of rapid ocular growth.²⁹ Between 28 weeks GA and term, retinal surface area doubles³⁰ and most of this growth takes place in the retinal periphery,³¹ where ROP develops. To summarize, there are regional neuroanatomical variations which are fully or partially offset by the eccentricity of the retinal vascular and neural organization. In our current dearth of knowledge of the preterm neonatal eye this aspect cannot be fully explored, but on the assumption that there is increased nasal retinal cell density, it is possible that this influences the early onset of ROP at this site. There is no obvious anatomical explanation for the sparing of the vertical, compared with the temporal, retina. Thus while neuroanatomical factors cannot be entirely excluded as factors in determining the location of ROP, their patterns are difficult to reconcile with the variations reported here.

The amount of light reaching the neonatal retina will now be considered. The retinal light dose (retinal irradiance) is influenced by several factors including the environmental illumination, eyelid-related factors, and ocular transmission. The illumination of neonatal units is typically constant throughout the 24-hour period, although in a few units this is reduced at night. Over the past two decades neonatal unit light intensities (illuminance) have increased by 5- to 10-fold, rising from about 100 lux³² to ~470 lux,^{33,34} with a peak of 10 000 lux where sunlight could contribute.³⁵ Lighting varies considerably in different sections of the neonatal unit.³⁴ Although the ROP study and our lighting surveys³⁴ were undertaken in the same neonatal units, they were not undertaken concurrently. The mean illuminance of our units was 470 lux (236 to 905 lux), ie, at the lower range quoted above, but this does not reflect the considerable variation between rooms within units and between units, as in certain intensive care areas of the nursery this rose to 905 lux (range 210 to 905 lux) whereas the mean illuminance of areas where older babies soon to be discharged to home are cared for is 233 lux (141 to 452 lux). The lighting characteristics are such that retinal irradiance of the preterm neonate is weighted toward the shorter wavelengths—the blue-light hazard range.³⁶ The amount of the environmental illumination reaching the eyelids has been investigated^{37,38} by taking recordings from above each eyebrow over the 24-hour period and later subjecting the data to time-series analysis—between 30% and 100% of environmental illumination reaches the eyelids (mean = 47%).^{37,38}

Preterm neonates, however, are usually nursed prone with the head turned to one side, and because of shielding of the lower eye by clothing, an arm, or the mattress, the lower eye/eyelid (eye closer to the mattress—see Fig 6) receives a mean of 25% (range 0% to 78%) of environmental light, whereas for the upper eye/eyelid (eye away from the mattress) this mean value is 86% (range 57% to 100%).^{37,38}

How much of the available lighting reaches the surface of the eye? This is determined by the duration of eyelid opening and transmission of light through the closed eyelids. Eyelid opening is a function of GA.³⁹ The eyelids of the neonate of 25 to 26 weeks GA are open less than 45% of the 24-hour period,



Fig 6. Neonate with head turned to one side, showing shielding of the lower eye.

although by 28 weeks PCA this has decreased to around 5% to 10%, increasing gradually over the ensuing weeks to 40% at 34 weeks PCA.³⁹ When the eye is closed, some light still reaches the cornea as the eyelid acts as a red-pass filter transmitting less than 14% of light in the adult^{40,41} and 21% in preterm neonates at 700 nm.⁴¹ In both groups this declines to less than 5% at ≤ 580 nm; further details of age changes are as yet unknown. No significant difference has been noted between ethnic groups in the nine neonates (white, Afrocaribbean, and Indo-Pakistani) tested so far, between 30 and 36 weeks PCA.⁴¹

Light has now reached the surface of the eye, having either passed by the open, or through the closed, eyelid. The ocular media of neonates are highly transmissive, especially in the blue and ultraviolet regions of the spectrum⁴² (the neonatal lens transmits $>90\%$ at ≤ 420 nm light).⁴³ In the most immature neonate, the cornea is hazy before 27 weeks GA and the transient vascular network surrounding the fetal lens (tunica vasculosa lentis) has not fully regressed until around 34 weeks GA.⁴⁴ The magnitude of this light-filter is unknown although it is likely to be slight, particularly after about 30 weeks GA. The retinal light dose is also modulated by pupil diameter. Before the onset of the pupillary light reflex between 30 and 34 weeks GA^{45,46} mean pupil diameter is 3.46 mm and after, 3.02 mm⁴⁶ (influenced by the level of environmental illumination), a value similar to that reported by Isenberg et al.⁴⁷ Light does not enter the eye exclusively through the pupil as the opaque coats (sclera and choroid) of the adult eyeball also transmit about 14% light,^{48,49} again predominantly at the red end of the spectrum (personal observation, J. R. and A. R. F., 1990). Using all the radiometric and physiological data available, the retinal irradiance received by a preterm neonate has been estimated and has been found to decrease with increasing PCA from $\sim 530 \mu\text{W}/\text{cm}^2$ at 24 weeks PCA to $\sim 240 \mu\text{W}/\text{cm}^2$ at 32 weeks PCA.^{37,38,50} Thus the neonate most at risk of developing severe ROP receives the highest light dose.

Optical radiation enters the interior of the eye by two routes, through the pupil and to a much lesser extent across the walls of the globe. The globe is not uniformly exposed to light, being partially shielded

superiorly and inferiorly by the upper and lower eyelids. Consequently those areas along the horizontal meridian and lying within the palpebral fissure receive the greatest light dose by transocular transmission. This has been confirmed by preliminary measurements obtained from neonatal postmortem material (personal observation, J. R. and A. R. F. 1990). On this basis, both the temporal and nasal retina should receive a similar light dose. However, the infant's head is generally turned to one side, with the lower eye receiving less light due to shielding by clothing, etc (Fig 6). The nasal retina of the upper eye, however, will receive more light as, when the eyelids are open (in contrast to diffuse retinal illumination occurring with eyelid closure), light will be directed predominantly onto this part of the retina.

In this article we have shown that ROP exhibits regional variations which cannot be fully explained by either vascular or neuroanatomical organization. Retinopathy commences in the most immature neonate, preferentially in the areas of the retina receiving the greatest light dose, and tends either to spare or develop later in the darker areas. Riley and Slater, in 1969⁷ and again in 1991,⁵¹ suggested that light may act as a factor in ROP by producing chemically excited states and generating free radicals. They considered that prolonged exposure to bright light, especially in association with raised oxygen levels, may initiate ocular damage and requested that the spectral characteristics of the neonatal environment be determined. This has now been undertaken (see Robinson and Fielder³⁶), but much remains to be investigated. Retinal blood flow and oxygen consumption are both higher in the dark than light in the human,^{52,53} although Hill and Houseman⁵⁴ recorded increased flow in some cats but decreased in others in response to dark exposure. Raised retinal oxygen tension has been measured in the light compared with the dark in the rabbit,⁵⁵ confirming decreased metabolic activity under these conditions. Recently, Dorey et al⁵⁶ demonstrated that blue light inhibits the growth of endothelial cells. These authors stated that since light or oxygen act synergistically with dopa to inhibit endothelial cell growth, exposure to light or oxygen would exacerbate the potential for vascular damage and hence the risk of ROP. Further consideration of the mechanism by which light may be a factor in the pathogenesis of retinopathy is beyond the scope of this article.

It is important to emphasize in this article that a causal relationship between light and ROP has not been demonstrated. However, this study has shown that there are regional variations in ROP and an understanding of the significance of its precise location may provide clues to the pathogenesis of this enigmatic condition. If it is eventually proven that light is significant in the development or contributes to the severity of ROP, here at least is one modality that can easily and immediately be controlled.

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